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Approved By
Western Hemisphere Department

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ASSESSING THE SCOPE FOR PERSONAL INCOME TAX REFORM AMID HIGH INFORMALITY[1]

Mexico's personal income tax (PIT) revenue remains low and informality high. This paper assesses the scope for mobilizing revenues through potential PIT reforms, taking into account the limits imposed by informality. Empirical estimates suggest that raising taxes would increase informality only moderately in the short run and, thus, could generate meaningful revenues. However, modelling work points to potentially more significant behavioral responses in the long run, once households, firms and prices adjust, highlighting the importance of considering general equilibrium effects when evaluating possible policy changes. Given prevailing levels of informality, the analysis suggests that reforms narrowly focused on top earners would improve progressivity but would need to be accompanied by formalization measures to raise meaningful revenue. On the other hand, broad rate increases would raise more revenue although increasing informality. Lowering pension-income exemptions and measures to foster labor force participation appear to face more benign trade-offs, with potential to generate significant revenue gains while supporting formality and progressivity. Overall, these results highlight that mobilizing revenues meaningfully will require a combination of tax reforms and greater efforts to foster formalization.

A. Introduction

1. Mexico's PIT revenue remains low relative to OECD peers, reflecting regressive tax expenditures, lower top tax rates, and moderate progressivity. Although PIT revenue has risen over time, reaching about 3.7 percent of GDP in 2023 and outpacing LA7 peers, it remains well below the OECD average (8 percent).[2] This reflects, in part, policy design. Tax expenditures stood at around 1 percent of GDP in 2025, with deductions and exemptions concentrated among higher-income taxpayers. Moreover, Mexico's top PIT threshold, relative to the average wage, is considerably higher and its top marginal rate lower than the OECD average. On the OECD's tax-wedge progressivity indicator, Mexico ranks 23rd among 38 OECD countries (OECD, 2026). At the same time, Mexico's PIT administration performs relatively well in some areas, with widespread electronic filing and strong on-time payment rates, suggesting limited scope for further revenue gains through PIT administrative improvements alone.

2. Informality remains high, underscoring the challenge of mobilizing PIT revenue. Informality has remained high compared to peers over the past decade and, while it is concentrated among low-income workers, it is also substantial for higher income deciles.[3] Disparities are also evident across education levels: 82 percent of workers with only primary education are informal,

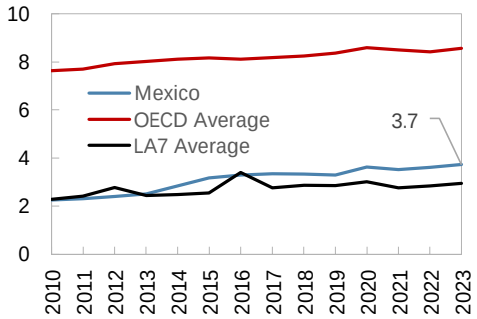
[1] Prepared by Lisa Kolovich, Vivian Malta, Joaquin Marandino Peregalli, Johanna Schauer, Monica Tran-Xuan, and Usama Zafar.

[2] Labor and self-employment income constitute an important part of the PIT base, suggesting that limited PIT revenue largely reflects limited mobilization from these sources.

[3] Informal employment is defined using the ENOE's main-job classification (*emp_ppal*): workers in the informal sector, subsistence-agriculture self-employed, unpaid family workers, and domestic, salaried, non-wage subordinate workers, and employers lacking social security coverage.

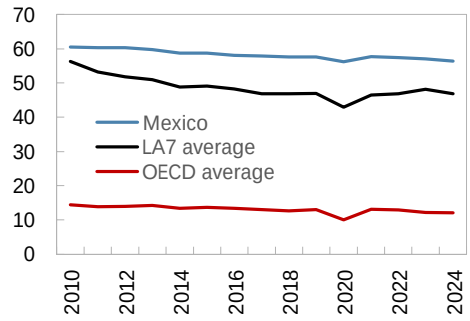
compared to 60 percent for secondary education and 30 percent for tertiary education. Gender gaps are more modest, with informality affecting 59 (57) percent of female (male) workers. These patterns highlight that potential PIT reforms aimed at boosting revenue would need to account for the impact of tax changes on formality-informality choices of different groups of workers. The tax burden alone, however, is unlikely to explain Mexico's persistently high informality, pointing to the need for a broader policy agenda to make formal employment more attractive and productive (Box 1).

Taxes on Income and Profit of Individuals
(Percent of GDP)



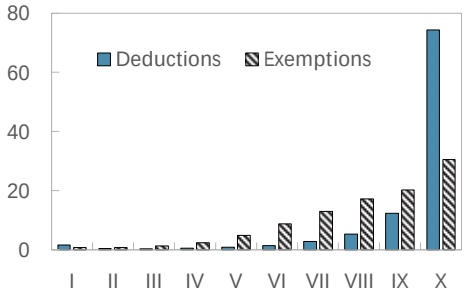
Source: OECD.

Informal Employment Rate
(Percent of employment)



Source: ILO.

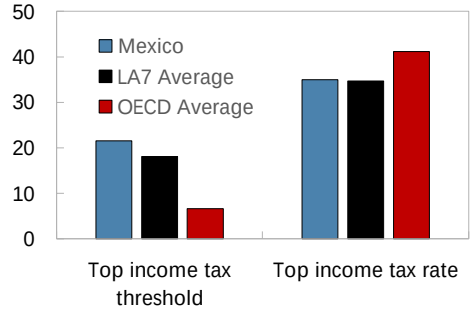
PIT Exemptions and Deductions by Income Decile
(Percent of total)



Source: Hacienda (2024); IMF staff calculations.

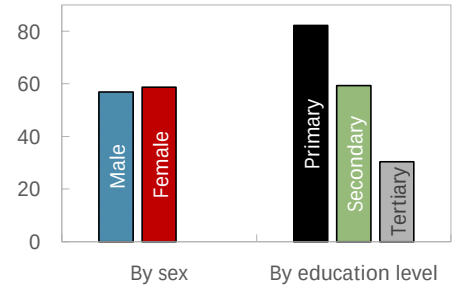
Top Income Tax Threshold and Rate

(Share of average wage)



Source: OECD.

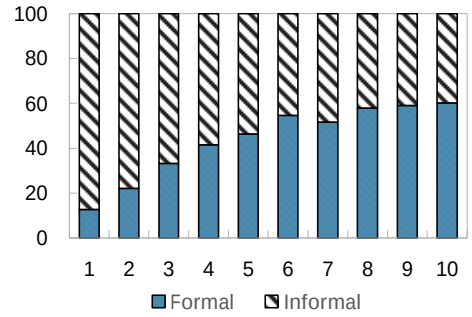
Informal Employment Rate by Sex and Education, 2025
(Percent of employed population)



Source: INEGI (ENOE), IMF staff estimates.

Formal Employment by Income Decile

(Percent of workers in decile, 2025)



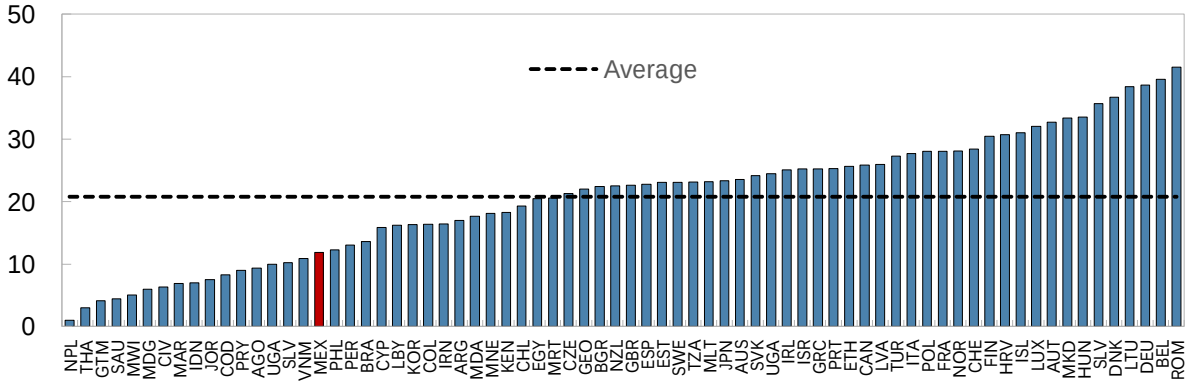
Source: ENOE.

Box 1. Mexico: How Mexico's Tax-Benefit System Shapes Work Incentives

The IMF's TaxFit model applies Mexico's tax and transfer rules to assess how taxes, social security contributions and cash transfers affect disposable incomes across household types and earnings levels. By comparing income under different earnings and employment scenarios, the model can identify where features of the tax-benefit system strengthen or weaken incentives to work, earn additional income, and participate in formal employment. TaxFit focuses on the direct effects of taxes and benefits on household incomes, thus providing a short-run, partial-equilibrium assessment of incentives.

Results indicate that the overall tax burden on labor income remains modest at lower earnings levels. The employment subsidy—a tax credit for low-income formal employees—offsets much of the personal income tax liability for workers earning close to the minimum wage, suggesting that moving from informal to formal employment leads to relatively minimal disincentive effects. Thus, while the tax-benefit system influences employment choices, tax incentives alone are unlikely to account for Mexico's high informality. After the removal of the subsidy, average effective tax rates rise only gradually as earnings increase, which contributes to Mexico's relatively low PIT revenue compared with OECD peers despite a progressive statutory PIT schedule (see chart).

Effective Tax Rate: Single Worker, No Children, Earns Average Wage
(Percent)

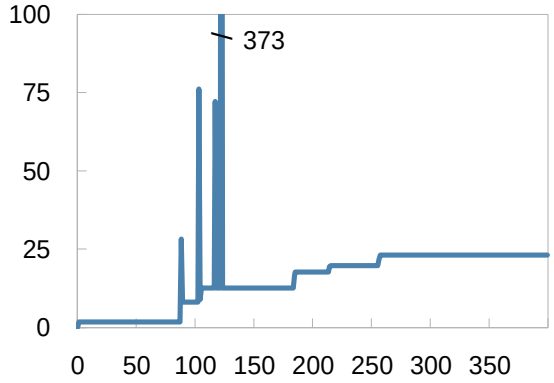


Source: IMF staff estimates.

The phasing out of the employment subsidy results in the tax burden increasing sharply. The structure of the tax-benefit system becomes more apparent when examining marginal effective tax rates (METRs), which measure how much of an additional peso of earnings is lost through higher taxes, social security contributions, or reduced benefits. Before 2025, workers faced several sharp increases in METRs around earnings levels close to the minimum wage, reflecting the stepped withdrawal of subsidy benefits (see chart). The 2025 reform substantially changed these incentives by eliminating several METR spikes while creating a single, more pronounced earnings threshold where the entirety of the employment subsidy is withdrawn (at about 120 percent of minimum wage). Still, a small increase in gross earnings at about 170 percent of minimum wage can result in a decline in disposable income, generating a METR that exceeds 800 percent. Although the distortion affects a relatively narrow range of workers, it highlights how the design of tax credits and subsidies can have consequences for work incentives that are not immediately apparent from statutory tax schedules alone. Overall, the TaxFit diagnostics show that design of tax credits and subsidies matters in shaping work incentives, particularly around key earnings thresholds.

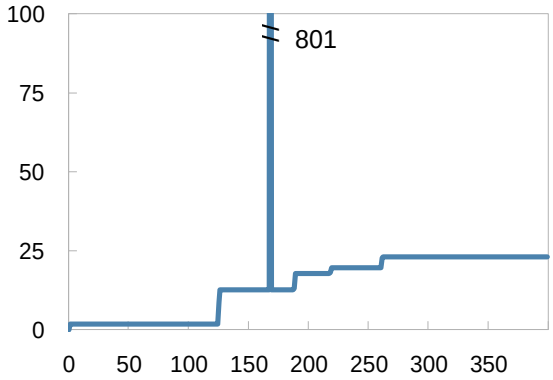
Box 1. Mexico: How Mexico's Tax-Benefit System Shapes Work Incentives (concluded)

Intensive METR (Taxes and Benefits): Single Worker, No Children (2024)
(X-axis: Percent of minimum wage; Y-axis: Percent)



Source: IMF staff estimates.

Intensive METR (Taxes and Benefits): Single Worker, No Children (2025)
(X-axis: Percent of minimum wage; Y-axis: Percent)



Source: IMF staff estimates.

B. Elasticity of Informality to Income Tax

3. The response of informality to changes in formal-sector taxes is central to assessing the revenue potential of PIT reforms. Higher PIT reduces the income return to formal employment and may induce workers to shift toward informal work, eroding the effective tax base. Using quarterly data from Mexico's National Survey of Occupation and Employment (ENOE) for 2005–25 and following the empirical approach in IMF (2024), the response of informality to changes in the formal–informal income gap across demographic groups can be estimated. The estimated elasticity can be combined with the average formal-sector wage in the sample to derive the implied short-run response of informality to an increase in the formal-sector tax burden (Annex I).

4. The results point to a moderate short-run response, concentrated among specific groups. On average, a tax increase equivalent to 1 percent of formal-sector wages would raise the informality rate by about 0.1 percentage points (Table 1). This effect works primarily through the female population and middle-skilled workers, while estimates for other groups are smaller or not statistically significant. The average effect is somewhat larger than in previous studies for other countries, which generally find insignificant or smaller responses (IMF, 2024; McKay et al., 2024).

Table 1. Mexico: Informality Elasticity w.r.t. Formal-Informal Income Gap, and Effect of 1% Tax on Informality

	(1)	(2)	(3)	(4)	(5)	(6)
	Full Sample	Male	Female	Primary	Secondary	Tertiary
Elasticity (at the sample mean)	-0.0391**	0.00317	-0.0635***	-0.00462	-0.0458***	0.00304
	(0.0153)	(0.0101)	(0.0169)	(0.00431)	(0.00994)	(0.0624)
Observations	2,490	1,245	1,245	830	830	830
1%-tax effect on informality (pp)	0.1	-0.01	0.2	0.04	0.2	-0.003

*** p<0.01, ** p<0.05, * p<0.1

Source: IMF staff calculations. Notes: Results are from a grouped estimator using sex × age group × education cells observed over quarters, weighted by cell size, with standard errors clustered at the cell level. Elasticity standard errors are computed via the delta method.

5. These estimates provide a short-run benchmark but do not capture the full long-run effects of PIT reform. The empirical approach identifies the average short-term historical response of informality to changes in the tax burden. It does not capture longer-term and reform-specific adjustments in labor supply and demand, or broader general-equilibrium effects. Section C presents a structural model that takes into account these adjustments and sheds light on long-run effects.

C. Policy Simulations

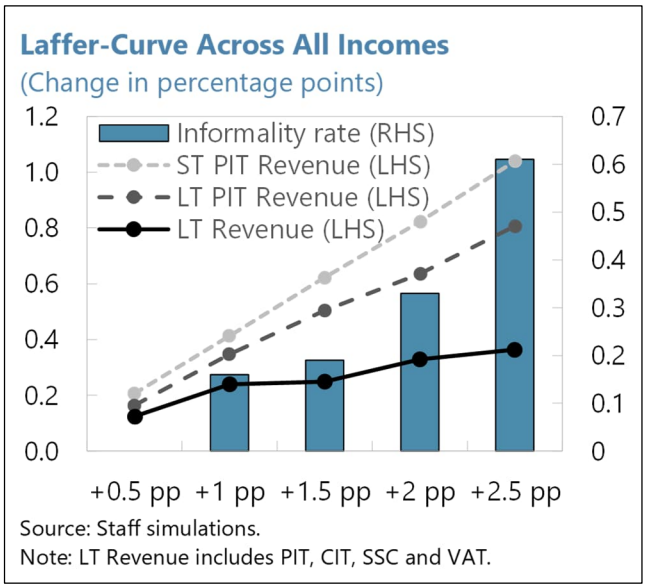
6. A general-equilibrium model with heterogeneous agents is calibrated to the Mexican economy. The model is a small open economy with overlapping generations, formal and informal labor markets, and agents that differ in productivity and gender. It is calibrated using, among other sources, the 2024 ENIGH's microdata and Mexico's tax and pensions systems to match the statutory 11-bracket income schedule (net of the employment credit), social-security contributions, and accrued pension rights (see further details in Annex II). The framework can trace how PIT and other reforms would affect labor supply decisions, prices, output, inequality, and government revenues in the long term. Three sets of reforms are considered: adjustments to PIT parameters for workers, reduction in pension tax expenditures, and policies that encourage female labor and formal employment.

7. The results highlight the importance of accounting for behavioral responses when evaluating revenue mobilization options. As informality in Mexico is pervasive and affects even high-income earners, reforms that focus solely on taxing high-income earners are able to reduce inequality but are not able to generate durable and meaningful revenues, as targeted workers shift some of their high-productive working hours to informality. On the other hand, expanding the PIT base can generate more durable revenue gains and still lower inequality in the long run, but would also have some impact on informality. Reducing tax subsidies on pensions earnings can boost tax collection with less labor-supply distortion, as future pensions earnings affect workers' labor supply with a discount factor. Moreover, reforms to improve formalization and facilitate female labor force participation can generate substantial revenues amid lower informality.

Labor Tax Reforms

8. Lowering the threshold for the top PIT bracket can raise progressivity but yields limited PIT revenues on its own. The first scenario lowers Mexico's top PIT threshold toward the OECD average: all labor income above MXN 1.13 million is taxed at the 35 percent marginal rate (instead of the current 32, 34 and 35 percent rates). In the short run (i.e., before accounting for general equilibrium responses), the reform would raise PIT revenues by 0.014 percent of GDP, reflecting the narrow tax base and informality among top earners.[4] In the long term, PIT gains from high-income earners would roughly be halved as the affected workers shift part of their working hours into informality. In general equilibrium, the reform also induces low- and middle-income workers — whose own tax rates are unchanged — to supply somewhat more hours, mostly to the formal sector, so that GDP is 0.3 percent higher and most of the additional revenue accrues through social security contributions. Revenues would rise although mostly through social security contributions. Overall, government revenues would rise by 0.09 percent of GDP: 0.05 from social security contributions, 0.02 from VAT, 0.01 from corporate income tax, and 0.01 from PIT. Given the labor-supply response of middle-income workers, the impact on aggregate informality would be limited, while the reform would be progressive, lowering the Gini coefficient by 0.4 points. Increasing the top tax rates (by up to 10 percentage points) would not deliver meaningful additional PIT revenue, as top earners would shift further into informality. These results point to limited scope for mobilizing revenue by taxing top earners alone, reflecting both the narrow tax base and the ability of high-income workers to shift activity into informality. Greater risk-based tax enforcement targeting these workers would therefore reduce their ability to shift hours into informality, improving the reform's revenue yield.

9. Broad PIT rate increases could boost revenues although with some negative impact on informality and reducing GDP. An illustrative Laffer curve is built by evaluating the impact on tax collection from raising marginal income tax rates across all 11 brackets: by 0.5 percentage points, then by 1.0, and so on up to 2.5 percentage points. In the short term (before accounting for general equilibrium responses), PIT revenue gains can reach more than 1 percent of GDP at the upper end (see chart). Over the long term, however, workers would shift hours into informality (and, for larger tax increases, also reduce total hours worked), lowering GDP. PIT collection would remain significant—retaining about 80 percent of the short-term gain—as all workers face the higher rates. However, collection of other taxes would

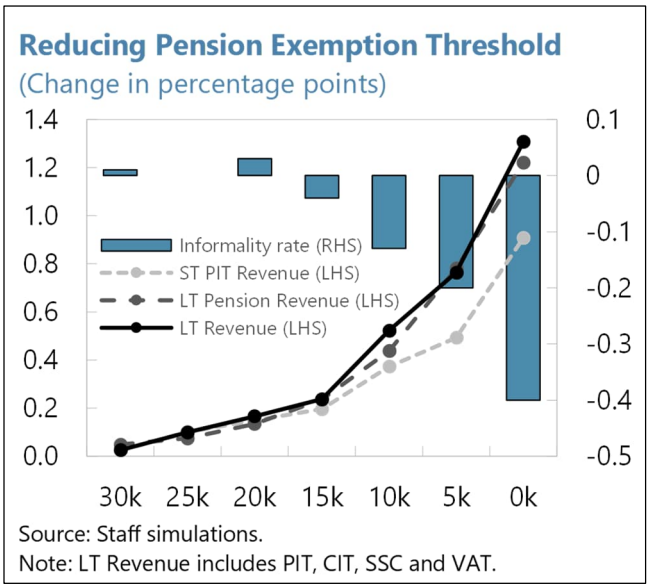


[4] The model is calibrated using ENIGH 2024 microdata, which indicates that only about 5 percent of gross formal labor income accrues above the MXN 1.13 million threshold, and that workers earning more than this threshold supply 29 percent of their working hours informally.

be eroded, as a result of lower output and higher informality, which would reduce VAT and social security collections. For example, for a 2.5 percentage point increase in PIT rates, long-run PIT revenues would rise by about 0.8 percent of GDP (compared to a 1.0 percent increase in the short run) but the overall revenue gain would be only 0.4 percent of GDP, amid a 0.6 percent increase in the informality rate. A reform like this would be progressive—the Gini coefficient would fall by up to 0.7 points—because the employment tax credits would shield the lowest brackets from the rate increase.

Tax Expenditure Reforms

10. Reducing the pension-income exemption could broaden the tax base and reduce income inequality. Given that Mexico's pension exemption is among the largest PIT tax expenditures, a simulation that lowers the 2024 exemption from MXN 53,000 to MXN 30,000, MXN 25,000, and progressively to zero is considered (see chart).[5] The model's long-run response highlights how shallow exemption cuts operate in an incentive-leakage regime: as the after-tax lifetime return to formal work falls, formal hours (and therefore earnings) drop. As a result, the pension-tax gain is largely offset by lower PIT and VAT collections. Deeper exemption cuts, on the other hand, trigger a wealth effect: as workers lose a larger part of their lifetime retirement income, they work more to restore their purchasing power. The additional hours go to the formal sector—informal work offers decreasing returns, whereas formal work still accrues (now more heavily taxed) pension rights — and the response is concentrated among men aged 50–64. Overall, revenue gains are positive for all exemption thresholds and rise, at an increasing pace, as the threshold declines. The impact on aggregate informality ranges from negligible to favorable. In addition, as pensions above the thresholds accrue to retirees in the upper half of the income distribution, all scenarios lower inequality, with the largest effect occurring at a threshold of MXN 10,000, where the Gini coefficient falls by 0.5 point. This reform is well suited to raising revenue with limited labor-supply distortions: because pension benefits accrue decades after the work decision, workers internalize the heavier pension taxation only in discounted form.



[5] Most mandatory pension contributions are not subject to tax. While the model does not capture this level of granularity, the actual proposal may need to be adjusted parametrically to avoid double taxation of the small share of pension contributions that may be taxed.

Formalization and Labor Market Policies[6]

11. Measures that make formal work more attractive affect mostly middle-income earners and can generate substantial revenue. In this simulation, agents' preferences for informal employment and for informal goods are reduced so that Mexico's informality rate drops by about 3 percentage points.[7] The scenario can be interpreted as a broad package of formalization reforms, including stronger enforcement of labor and tax regulations, simplified compliance procedures, and greater use of digital payments and invoicing, and information sharing across tax and other administrative databases. Among men, the response largely reflects a shift from informal to formal hours, concentrated among middle-income workers aged 35–64, while younger men increase formal hours without reducing informal work. Among women, formal hours also rise and informal hours fall, but female labor force participation declines by 2.6 percentage points. In particular, for middle-income women, lowering the incentive for informal work pushes some out of the labor force rather than into formal jobs. For these women, participation is a threshold decision: informal work—with its flexibility and non-pecuniary value—was what made working worthwhile relative to home production, whereas the formal alternative did not clear that threshold; once the informal option becomes less attractive, staying home dominates both. The Gini coefficient declines by one point, as middle-income workers shift to formal work, where earnings are higher. In this scenario, GDP rises by 2.7 percent, and total revenue by 1.2 percent of GDP in the long run, led by social security contributions (0.5 percent of GDP) and PIT (0.3 percent of GDP).[8]

12. Measures that raise female labor force participation can also raise revenue while reducing informality. This scenario lowers the utility cost of married women working outside the home, capturing a broad range of possible policies, including childcare support, safer transportation, and improved education and training opportunities.[9] As in the previous simulation, the size of the shock is calibrated to a target—here, a 3 percent increase in formal GDP. Under this scenario, female labor force participation rises by about 5 percentage points as married women enter the labor market. At the same time, higher household income and a lower price of informally produced goods lead their husbands to slightly reduce informal hours. GDP rises by 2.5 percent and revenue by about 0.7 percent of GDP in the long run, driven primarily by higher social security contributions

[6] Unlike the tax and pension scenarios, which change statutory parameters, the two scenarios below are calibrated to reach a target outcome — a 3 percentage point reduction in informality and a 3 percent increase in formal GDP, respectively — so their fiscal effects are conditional on those outcomes being achieved rather than yields of specific policy measures.

[7] This can represent simpler compliance procedures and stronger enforcement (ILO, 2014) as well as wider adoption of digital payments (COFECE, 2025) or further leveraging of electronic invoicing that could increase transaction traceability and strengthen tax compliance (CIAT, 2018).

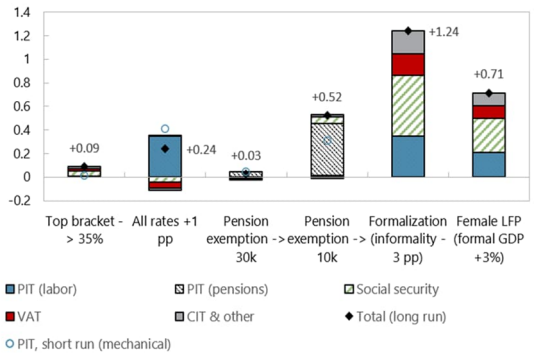
[8] The model's underprediction of formal hours at one to two minimum wages (Annex II, Table 3) biases the labor-income-tax component of these gains downward by at most a few hundredths of a percent of GDP; the social-security and VAT components are unaffected.

[9] Studies on Mexico labor policies emphasize expanding affordable childcare and early childhood education, increasing access to full-day primary schools, and easing elder-care constraints to support women's employment (OECD, 2024, 2026; Talamas Marcos, 2022).

(0.3 percent of GDP) and PIT revenue (0.2 percent of GDP). Informality declines by 0.7 percentage points, and the Gini coefficient falls modestly, by 0.1 point.

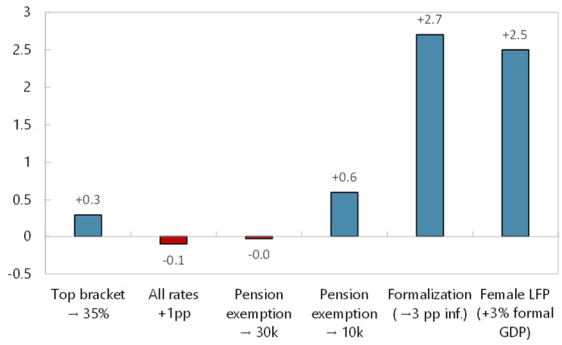
Mexico: Long-Run Effects from Potential PIT and Labor Market Reform

Revenue Gain by Instrument, Long Run
(pp of GDP)



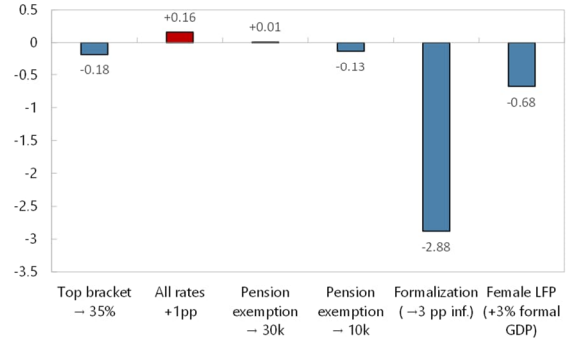
Source: Staff estimates.

Real GDP
(Percent of GDP)



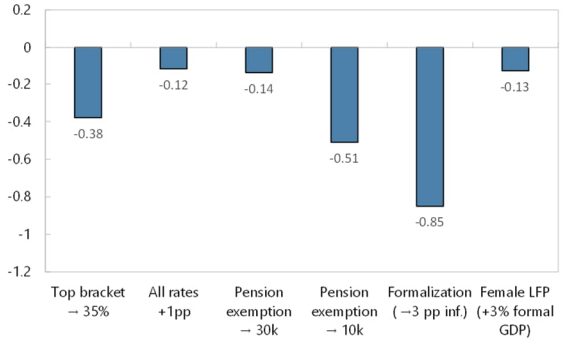
Source: Staff estimates.

Informality Gap
(pp of total hours)



Source: Staff estimates.

Gini Coefficient
(Points)



Source: Staff estimates.

D. Conclusions

Mexico has scope to raise PIT revenue although the size of the tax base and informality can be important constraints. While empirical estimates suggest that a higher formal-sector tax burden would only moderately increase informality in the short run, a structural model points to potentially more significant behavioral responses over time, highlighting the constraints posed by informality. Lowering the threshold for the top PIT bracket could improve progressivity but would need to be accompanied by formalization measures to raise meaningful revenue. Broader tax rate increases would raise more revenue although also raising informality. Reducing tax expenditures on pensions or supporting formalization and female labor force participation face more benign trade-offs, with potential to generate significant revenue gains while supporting progressivity. Stronger risk-based enforcement, digitalization, and information sharing could complement these reforms by limiting underreporting and making revenue gains more durable. Overall, these results highlight that

mobilizing revenues meaningfully will require a combination of tax rate increases and greater efforts to foster formalization.

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Annex I. Elasticity Estimation

This annex presents the empirical approach used to estimate how informality responds to changes in the income return to formal employment. The results provide a reduced-form measure of this response by exploiting variation across demographic groups and over time.

1. The analysis follows the grouped-estimator tradition in the labor-supply literature. This includes Jäntti et al. (2015), McKay et al. (2024), and IMF (2024), building on Blundell et al. (1998). The estimator relates changes in informality across demographic cells and over time to variation in the formal–informal income gap, while controlling for persistent differences across groups and common time shocks.

2. The baseline specification relates cell-level informality to the formal–informal income gap. The estimating equation is:

$$P(y_n > 0)_{it} = \alpha + \beta(x_f - x_n)_{it} + \mu_i + \lambda_t + \eta_{it}$$

where:

- $P(y_n > 0)_{it}$ is the cell informality rate for cell i (sex × age × education) at quarter t
- $(x_f - x_n)_{it}$ is the mean formal–informal income gap
- μ_i are cell fixed effects
- λ_t are quarter fixed effects.

3. The specification is estimated using weighted ordinary least squares. Regressions are weighted by cell size, and standard errors are clustered at the cell level. Cell fixed effects absorb time-invariant differences in informality across demographic groups, while quarter fixed effects control for aggregate shocks common to all groups.

4. The estimated semi-elasticity is converted into an elasticity evaluated at sample means. The elasticity is calculated as follows:

$$\varepsilon = \hat{\beta} \times \frac{\overline{(x_f - x_n)}}{\overline{P(y_n > 0)}}$$

where

- $\hat{\beta}$ is the estimated coefficient from the grouped regression
- $\overline{(x_f - x_n)}$ is the sample mean of the formal-informal income gap
- $\overline{P(y_n > 0)}$ is the sample mean informality rate

The elasticity measures the percent change in the informality rate associated with a one percent change in the formal–informal income gap, evaluated at the sample means. A negative elasticity indicates that a larger income return to formal employment is associated with lower informality, while a positive elasticity indicates the opposite.

The estimated elasticity is used to calculate the change in informality associated with a 1% income tax increase. A tax increase equivalent to 1 percent of average formal-sector income reduces the formal–informal income gap by $0.01\bar{x}^f$, implying that the associated change in the average informality rate is

$$\Delta\bar{P} = -\beta(0.01\bar{x}^f).$$

Table 1 reports this effect as the percentage-point change in the informality rate.

5. Identification comes from differential changes in the formal–informal income gap across demographic cells over time. The estimator uses within-cell variation after controlling for permanent group differences and common quarter effects, making it well suited to repeated cross-section data. The estimates should be interpreted as conditional associations, as changes in cell-level income gaps may also reflect changes in worker composition within cells over time.

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Annex II. Description of the General Equilibrium Model

This annex presents the overlapping-generations general-equilibrium model used in section C of the paper.[1]

Overview

1. The model is a small open economy populated by overlapping generations of heterogeneous households. Agents live for $J = 13$ periods of five years each, entering the economy at age 20 and dying at 84; they may work during the first 9 periods (ages 20 to 64) and are retired thereafter. The economy has two sectors: a perfectly competitive formal sector that produces a tradable good with capital and labor, and an informal sector in which individuals produce a non-tradable good on their own account, using labor only and with decreasing returns to scale. Agents differ by generation, gender, household type (married couples, single men, and single women), years of schooling, and an idiosyncratic productivity endowment ε drawn from a distribution with a lognormal body and a Pareto upper tail.

Households' Problem

2. Married couples jointly maximize lifetime utility. Husband and wife belong to the same generation and are assortatively matched, sharing the same endowment rank. A couple with endowment ε chooses consumption of the formal good and of the informal good, the hours each spouse supplies to the formal and the informal sector, and savings, so as to maximize

$$\max \sum_{t=1}^{J} \beta^{t-1} \Big[\zeta_{fo} \log c_t^{fo} + \zeta_{inf}^{eff} \log c_t^{inf} - \Psi_{m,t}(l_t^m)^{\chi} - \Psi_{f,\varepsilon,t}\big(l_t^f\big)^{\chi} + A_{m,\varepsilon}\xi\big(l_t^{m,inf}\big) + A_{f,\varepsilon}\xi\big(l_t^{f,inf}\big) - Q_{\varepsilon}\mathbf{1}\{l_t^f > 0\} \Big]$$

where β is the discount factor; $\zeta_{fo} = 1$ and ζ_{inf}^{eff} are the preference weights on the formal and the informal goods (ζ_{inf}^{eff} is non-homothetic); $\Psi_{i,t}(.)$ scales the disutility of work, with χ governing the curvature of labor disutility; and $l_i = l_i^{fo} + l_i^{inf} \in [0,1]$ is the total time worked by spouse i. Informal work carries a non-pecuniary amenity: the term $A_{i,\varepsilon}\xi(.)$ values the flexibility and autonomy of own-account work as a proportional premium on the informal return. [2]Married women face a fixed utility cost of participating, $Q_{\varepsilon} = max\{q_0 + q_1 ln\varepsilon, 0\}$ with $q_1 > 0$, incurred in any period in which they supply positive labor hours. This "care wall" is a reduced-form representation of unpaid care and domestic work.

[1] Model developed by Vivian Malta (SPR). This model is an extension of the D-GEN models built inside the IMF (general equilibrium OLG models that analyze fiscal and structural policies), and it builds on the models described in Kolovich, MacDonald and Malta (2026) and Malta et al (2019).

[2] Operationally, the amenity enters agents' hours-allocation condition as a wedge that multiplies the marginal return from working informally by $(1 + A_{i,\varepsilon})$, with $A_{i,\varepsilon} = A_i \varepsilon^s$ and s a small positive productivity gradient (Table 2).

3. The couple's period budget constraint is

$$(1+\tau_c)c_t^{fo} + p_{inf}c_t^{inf} + a_t = \sum_{i\in\{m,f\}}\left[y_{t,i}^{fo}(1-\tau_{ss}) - T\left(y_{t,i}^{fo}\right) + p_{inf}y_{t,i}^{inf} + Pen_{t,i} - T(Pen_{t,i})\right] + Tr_\varepsilon + a_{t-1}[1+r^*(1-\tau_a)]$$

where τ_c is the value-added tax, levied on the formal good only; $p_{inf}$is the price of the informal good; τ_{ss} is the employee social-security contribution, applied to formal earnings up to the annual cap of 25 UMA; $T(.)$ is the personal income tax (ISR), computed from the statutory 11-bracket schedule of marginal rates and fixed quotas, net of the employment credit (*subsidio al empleo*) for low earners; Tr_ε is a targeted cash transfer to households in the lowest productivity types; $Pen_{t,i}$ is pension income, received only in retirement and taxed under the ISR schedule on the amount exceeding the pension exemption; r* is the international real interest rate; τ_a is the tax on savings income.

4. Gross formal earnings and informal output of spouse $\boldsymbol{i}$ are

$$y_{t,i}^{fo} = w_{fo}\phi_{fo}^i h_{t,i} l_t^{i,fo}, \quad y_{t,i}^{inf} = \phi_{inf}^i \left(h_{t,i} l_t^{i,inf}\right)^{\alpha_{inf}}$$

where w_{fo} is the formal wage per efficiency unit; $h_{t,i}$ is human capital; $\alpha_{inf} < 1$ governs decreasing returns in informal production; and $\phi_{inf}^i, \phi_{fo}^i \leq 1$ capture residual gender gaps in efficiency in each sector, normalized to one for men.

5. Human capital depends on the endowment, schooling, and experience. In the first period, $h_{1,i} = \varepsilon e_i(\varepsilon)^{\alpha_e}$, where $e_i(\varepsilon)$ is years of schooling (calibrated exogenously using ENIGH 2024 microdata). Thereafter and until retirement, male human capital grows at exogenous age-specific rates $g_{m,t-1}$ estimated from Mincer regressions, while female human capital grows at $g_{f,t-1}$ only if the woman worked in the previous period and is stagnant otherwise, capturing the scarring effect of career interruptions: $h_{t,m} = \left(1+g_{m,t-1}\right)h_{t-1,m}, \quad h_{t,f} = \left(1+g_{f,t-1}\mathbf{1}\{l_{t-1}^f > 0\}\right)h_{t-1,f}$, $for\ t = 2, \ldots, 9$.

6. Pensions are the key life-cycle return to formal work. Following the rules of the IMSS system, a worker who contributed for at least four periods (20 years) receives a contributory pension based on the final formal wage, up to the contribution cap; all retirees also receive the non-contributory universal pension (Pensión para el Bienestar), modeled as a flat benefit p. Because the pension depends on the worker's own contribution history, its present value is endogenous, and when choosing formal hours agents value the pension rights accrued on an extra unit of formal earnings. The accrual rate is disciplined so that a benchmark worker's accrued rights reproduce the net-of-tax pension implied by the IMSS formula.

7. Single households solve the same problem without the spouse's terms and without utility cost Q_ε.

Production

8. The formal good is produced by a representative competitive firm with a Cobb-Douglas technology in capital and effective labor. The firm pays a levy τ_v on revenue (calibrated to reproduce corporate income tax collections), employer social-security contributions τ_{ss2} on its

wage bill up to the cap, and rents capital at the cost $r^{fo} = r^* - s_k$, where s_k is a small wedge between the households' return on savings and the firms' cost of capital, calibrated to match Mexico's capital-to-GDP ratio:

$$\max (1-\tau_v)\, K^{\alpha_k}\left(L_m^{fo} + \phi_{fo} L_f^{fo}\right)^{1-\alpha_k} - (1+\tau_{ss2}) w_{fo}\left(L_m^{fo} + \phi_{fo} L_f^{fo}\right) - \left(r^{fo} + \delta_k\right) K$$

The informal good is produced on own account with output $y_{t,i}^{inf} = \phi_{inf}^{i}\left(h_{t,i} l_t^{i,inf}\right)^{\alpha_{inf}}$.

Government and Equilibrium

9. The government collects the personal income tax (net of the employment credit), the tax on pension income above the exemption, employee and employer social-security contributions, the corporate levy, the tax on savings income, and the value-added tax; it pays contributory pensions, the universal pension, and the targeted transfer. Given the fiscal parameters and the world interest rate, a stationary competitive equilibrium consists of household allocations for every household type, endowment, and age; a production plan for the formal firm; and a price system — the formal good as numéraire, the informal price p_{inf}, the formal wage w_{fo}, and the cost of capital r^{fo} — such that households maximize utility, the firm maximizes profits, and markets clear. The informal goods market clears domestically:

$$\sum_{e,i,j} \mu_{e,i,j}\, c_{e,i,j}^{inf} = \sum_{e,i,j} \mu_{e,i,j}\, \phi_{inf}^{i}\left(h_{e,i,j} l_{e,i,j}^{inf}\right)^{\alpha_{inf}}$$

which determines p_{inf}; the formal good is tradable, so any excess demand is absorbed by the external account; effective formal labor equals the sum of hours supplied to the firm weighted by human capital and the gender efficiency term; and the capital stock follows from the firm's first-order condition.

Calibration

10. The model is calibrated to 2024 using the ENIGH household survey, administrative earnings statistics, national accounts, and the parameters of Mexico's tax and pension systems. Exogenously set parameters are listed in Table 1. Demographics and household composition come from ENIGH (74 percent of adults live in couples, 13 percent are single men, and 12 percent single women; the age distribution determines generation weights). Years of schooling by gender are assigned to the 21 endowment types in ascending order, ranging from 7.0 to 16.2 years for men and from 6.8 to 15.8 for women. Returns to experience are estimated by Mincer regressions on ENIGH and decline with age. The residual gender gaps in efficiency are calculated using Oaxaca-Blinder decomposition (11 percent in the formal sector and 25 percent in the informal sector), and the informal returns-to-scale parameter $\alpha_{inf} = 0.62$ is estimated directly from the relationship between informal earnings and hours in ENIGH.

11. Tax parameters replicate the statutory ISR schedule bracket by bracket, including the fixed quotas and the employment-credit bands, expressed in model units through the reference income MW — the full-time formal earnings of a man with the lowest endowment, 90,859 pesos per year in the data (minimum wage). Social-security contributions use effective

rather than statutory rates (2.8 percent employee, 20.4 percent employer) with the 25-UMA cap; the VAT and the tax on savings income are also effective rates; and the corporate levy is scaled so that baseline collections match corporate-tax receipts. Pension rules replicate the IMSS eligibility threshold, the contribution cap, and the 2024 exemption of pension income up to 636,000 pesos per year (53,000 per month); the accrual rate that workers attach to marginal formal earnings is not a free parameter but is computed within the model, as the benchmark worker's net-of-tax IMSS-formula pension divided by the present value of that worker's gross formal earnings, so that accrued rights are consistent with the pension formula and with pension taxation; and the universal pension is set at 40 percent of the reference income.

Table 1. Mexico: List of Exogenous Parameters

Parameter	Description	Value	Source
J	Number of periods	13 (five years each; adults from 20 to 84)	Model structure
T_l	Length of working life	9 periods (retirement at age 65)	Model structure
μ_1, μ_2, μ_3	Share of couples, single men, single women	0.744, 0.134, 0.121	IMF staff calculations using ENIGH 2024
n	Number of endowment types	21 (18 equal-mass bins to p95; Pareto bins p95–99, 99–99.9, top 0.1%)	Model structure; Pareto coefficient from density continuity at p95
$e_m(\varepsilon)$, $e_f(\varepsilon)$	Years of schooling by gender and income level	Men 7.0–16.2; women 6.8–15.8	IMF staff calculations using ENIGH 2024
α_e	Elasticity of human capital to schooling	0.312	IMF staff calculations using ENIGH 2024
g_m, g_f	Returns to experience (Mincer, by age group and gender)	Men 0 to 0.093; women 0 to 0.081 per period	IMF staff calculations using ENIGH 2024
$\hat{H}m$, $\hat{H}f$	Age productivity profiles	Hump-shaped, 0.85–1.10 (men), 0.75–1.25 (women)	IMF staff calculations using ENIGH 2024
β	Discount factor	0.94 per year	Standard value
r*	Real interest rate on savings	3.0 percent per year	Mexico's long-run real rate in the latest Banxico report
α_k	Capital share	0.33	National accounts / standard value
δ_k	Depreciation rate of capital	5 percent per year	Standard value
α_{inf}	Returns to scale, informal sector	0.616	IMF staff calculations using ENIGH 2024 (informal earnings-hours regression).
ϕ_{fo}, ϕ_{inf}	Gender pay gap, formal / informal	0.886, 0.755	IMF staff calculations using ENIGH 2024 (using Oaxaca-Blinder decomposition)
τ_c	Value-added tax (effective)	0.105 (statutory 0.16)	SHCP revenue statistics / IMF staff calculations

Table 1. Mexico: List of Exogenous Parameters (concluded)

Parameter	Description	Value	Source
τ_v	Corporate levy on formal revenue	0.035 (statutory 0.30 on profits)	SHCP revenue statistics / IMF staff calculations
τ_{ss}, τ_{ss2}	Social security, employee / employer (effective)	0.028 / 0.204, on earnings up to 25 UMA (990,701 pesos)	IMSS, IMF staff calculations
τ_a	Tax on savings income	0.10	SHCP / IMF staff calculations
MW	Reference income	Full-time formal earnings of a type-1 man (MXN 90,859 per year)	Minimum wage in Mexico in 2024
T(.)	Income tax function	Statutory 2024 schedule: 11 brackets, rates 1.92–35 percent, with fixed quotas; employment credit of up to MXN 4,884 phased out at 88,588	LISR 2024 (SAT), statutory schedule and *subsidio al empleo*
ρ	Pension accrual (marginal valuation of formal earnings)	Computed within the model: net benchmark IMSS pension ÷ PV of benchmark gross formal earnings (≈2 percent per year).	IMF staff calculations using IMSS rules
—	Pension eligibility	Contributory pension after 20 years of contributions, based on the final formal wage up to the cap	IMSS rules
—	Pension exemption	MXN 636,000 per year (53,000 per month)	LISR 2024
p	Universal pension *(Pensión para el Bienestar de las Personas Adultas Mayores)*	40 percent of MW	Program rules 2024, IMF staff calculations
Tr	Targeted transfer (*Beca Benito Juárez)*	10 percent of MW for the lowest four endowment types	Beca Benito Juárez 2024 rules and IMF staff calculations
Sc	Equivalence scale, married households	3.73 (average household size)	ENIGH 2024, average household size
var(me)	Measurement-error variance in log earnings	0.206	GRID database (uses IMSS administrative records)

The remaining parameters are calibrated endogenously and jointly so that the simulated baseline reproduces empirical moments of the Mexican economy. Table 2 reports the calibrated values and Table 3 the fit; the weighted mean absolute percentage error across the fifteen targets is 3.7 percent.

Table 2. Mexico: List of Endogenous Parameters

Parameter	Description	Calibrated value
σ	Standard deviation of log endowments (lognormal body)	0.541
χ	Labor supply curvature (Frisch elasticity $1/(\chi-1)$)	1.737
Ψ_m	Disutility of labor, men	0.759
Ψ_f	Disutility of labor, women (level)	1.088
ψ	Disutility of labor, women (endowment gradient)	−1.897
q0	Care wall, level	0.227
q1	Care wall, endowment gradient (on ln ε)	0.116
A_m	Informal amenity, men	0.839
A_f	Informal amenity, women	1.460
s	Informal amenity, productivity gradient	0.079
ζ_{inf}	Informal preference scalar	0.505
s_k	Savings-return / cost-of-capital wedge	−0.007 (firms' cost of capital 3.7 percent)

Source: IMF staff calculations.

Table 3. Mexico: Model Fit — Empirical Targets Versus Simulated Baseline
(In percent, unless noted)

Target statistic	Data	Model
Formal share of GDP	74.6	76.7
Formal share of hours (FTE)	46.8	45.9
Formal/informal hourly wage premium (ratio)	1.73	1.73
Male labor force participation (hours per capita)	73.7	74.2
Female labor force participation (headcount)	61.7	61.0
Female/male relative hours (ratio)	1.11	1.11
Female LFP, poorest quintile	49.1	49.3
Female LFP, richest quintile	73.2	70.9
Informal share of consumption	37.0	33.6
Variance of log labor earnings	1.18	1.18
Capital-to-GDP ratio	2.80	2.80
Informal share of hours, top 5 percent of earners	28.9	29.0
Informal share of hours, men	53.6	54.6
Informal share of hours, women	52.6	53.1
Formal hours in income-tax brackets 3–5 (share)	51.6	34.1[1]

Sources: ENIGH 2024; national accounts; GRID; and IMF staff calculations.
Note: the earnings-variance target is matched after adding a measurement-error component (variance 0.206) to model earnings, consistent with administrative earnings statistics.

[1] The model underpredicts the share of formal hours in income-tax brackets 3–5 (34.1 versus 51.6 percent). These brackets cover earnings between one and two minimum wages, where observed formal wages concentrate because Mexico's minimum wage sets a floor on formal pay. The model's continuous productivity distribution does not generate this concentration and assigns part of the formal workforce to earnings below the minimum wage. The discrepancy affects only the labor income tax, whose effective rate is higher at the minimum wage (6.7 percent) than below it (3.6 percent); applying the floor to the model's earnings would raise baseline labor-ISR collections by about 3 percent. The results for the top bracket and the pension exemption are unaffected, as their bases are matched to the data, and so are social-security contributions and VAT, which are proportional to earnings and consumption. The ISR component of the formalization scenarios is therefore understated, by at most a few hundredths of a percent of GDP. Introducing a binding minimum wage in the formal sector is left for future work.

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